MILITARY SPECIFICATION SHEET

PIN-RIVET, THREADED, TYPE V (100° FLUSH MS24694 TENSION HEAD), CLASS 1 (7075 ALUMINUM, 45 KSI Fsu AND 75 KSI Ftu)

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the pin-rivet, threaded described herein shall consist of this document and the issue in effect of MIL-P-87988.

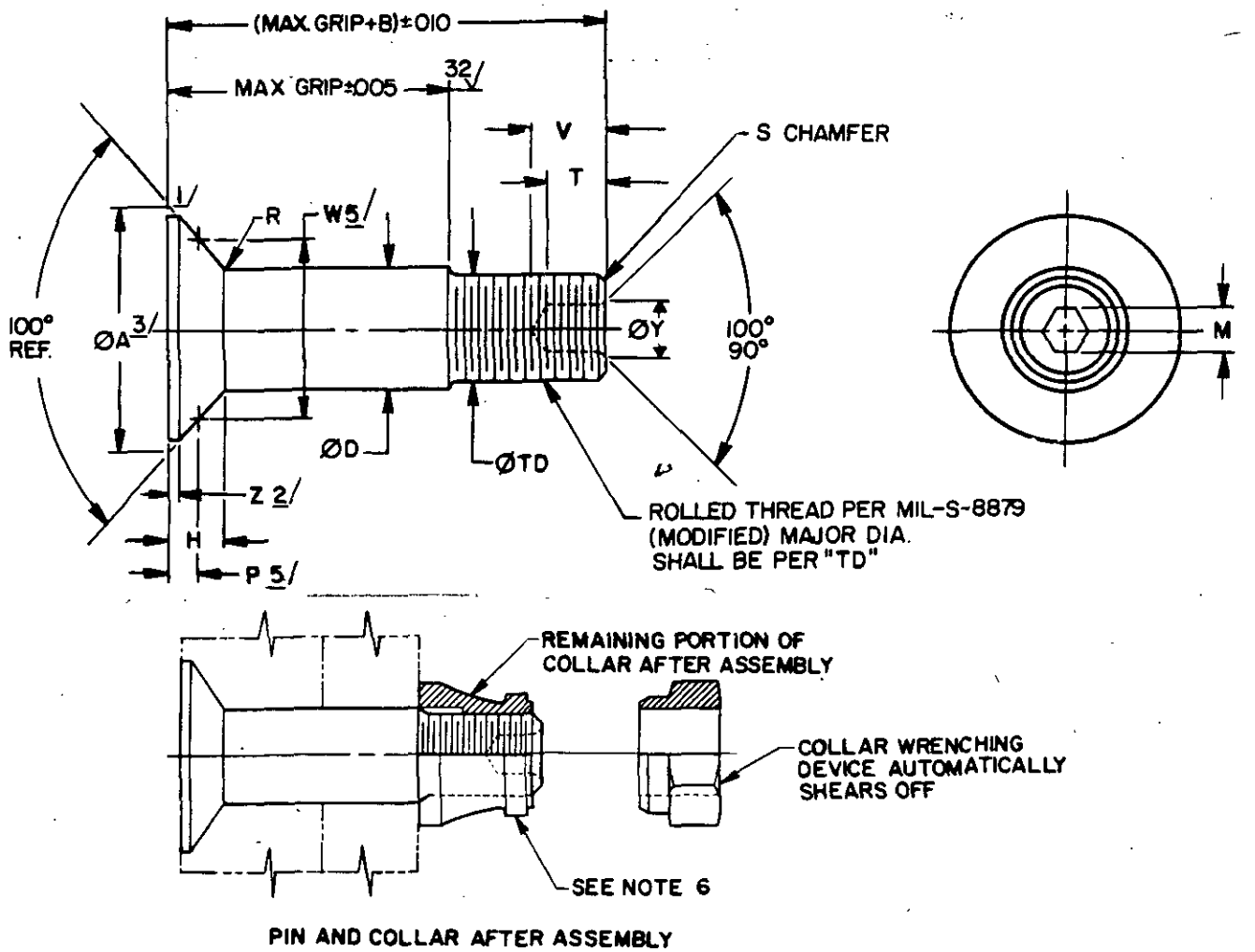


FIGURE 1. PIN-RIVET, THREADED, TYPE V (100° FLUSH MS24694 TENSION HEAD), CLASS 1 (7075 ALUMINUM, 45 KSI Fsu AND 75 KSI Ftu)

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NOTES:

1. Dimensions in inches to be met after application of final coating or surface finish, unless otherwise specified.
1/ 2. Head edge out of roundness shall not exceed "F".
3. Concentricity: conical surface of head to "D" diameter within .005 FIM.
4. "H" dimensioned from maximum "D" diameter.
2/ 5. Curved or flat edge manufacturer's option.
6. Parts identified with HI-Shear HL29 or M87988/3 are interchangeable in accordance with TABLE III.
7. Hole preparation in accordance with NAS618. (Information only)
4/ 8. Evidence of broken edge across points.
3/ 9. Concavity of head surface shall not exceed .001. Convexity of head surface shall not exceed .003 for -5, -6 and -8; .004 for -10 and -12; .006 for -14 and above (not including head marking identification).
5/ 10. All head dimensions shall comply with requirements of TABLE I of this slash sheet and TABLE X of procurement specification.

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TABLE I. DASH NUMBERS AND DIMENSIONS

DIA NO.	NOM SIZE	THREAD (MODIFIED)	ØA	B REF	ØD	ØTD	1/ F	H	R RAD	2/ Z MAX	S CHAMFER REF	SOCKET				DOUBLE SHEAR POUNDS MIN	TENSION POUNDS MIN
												M HEX	T DEPTH	V MAX DRILL DEPTH	ØY		
5	5/32	.1640-32 UNJC-3A	.3304 .3256	.312	.1635 .1625	.1595 .1570	.004	.0700 .0680	.025 .015	.015	.031 x 45°	.0645 .0635	.135 .115	.165	4/	1,760	1,030
6	3/16	.1900-32 UNJF-3A	.3813 .3765	.325	.1895 .1885	.1840 .1810	.005	.0805 .0785	.030 .020	.015	.031 x 45°	.0806 .0791	.135 .115	.175	.119 .104	2,540	1,640
8	1/4	.2500-28 UNJF-3A	.5066 .5018	.395	.2495 .2485	.2440 .2410	.006	.1079 .1059	.030 .020	.015	.031 x 45°	.0967 .0947	.150 .130	.200	.142 .122	4,500	2,950
10	5/16	.3125-24 UNJF-3A	.6335 .6287	.500	.3120 .3110	.3060 .3020	.007	.1349 .1329	.040 .030	.015	.047 x 45°	.1295 .1270	.170 .150	.240	.180 .160	7,050	4,660
12	3/8	.3750-24 UNJF-3A	.7604 .7556	.545	.3745 .3735	.3680 .3640	.008	.1619 .1599	.040 .030	.015	.047 x 45°	.1617 .1582	.200 .180	.279	.217 .197	10,100	7,200
14	7/16	.4375-20 UNJF-3A	.8884 .8812	.635	.4370 .4360	.4310 .4266	.009	.1895 .1865	.050 .040	.022	.047 x 45°	.1930 .1895	.230 .210	.318	.253 .233	13,800	9,350
16	1/2	.5000-20 UNJF-3A	1.0139 1.0068	.685	.4995 .4985	.4930 .4880	.010	.2160 .2130	.050 .040	.022	.047 X 45°	.2242 .2207	.260 .240	.363	.289 .269	16,000	13,000

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REQUIREMENTS:

1. Material. 7075 Aluminum alloy (UNS A97075) in accordance with QQ-A-225/9 or QQ-A-430.

2. Protective coating and surface treatment

   All pins shall be anodized in accordance with MIL-A-8625 Type II, Class 1.

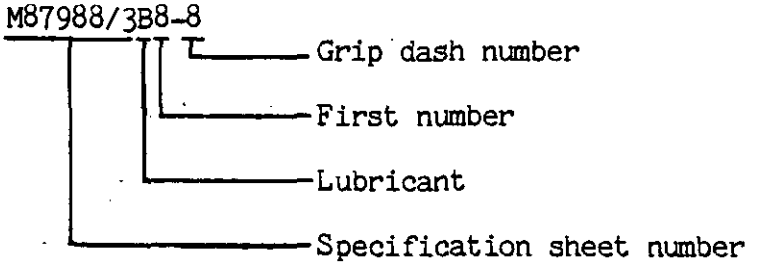
   Lubricant:
   B - Solid film lubricant in accordance with MIL-L-46010 Type I.
   C - Cetyl alcohol lubricant in accordance with MIL-L-87132 Type I, Grade B.

3. Heat treatment. Age to T6 condition in accordance with MIL-H-6088.

4. Surface texture. Unless otherwise specified, all surfaces shall have a 125 microinch finish.

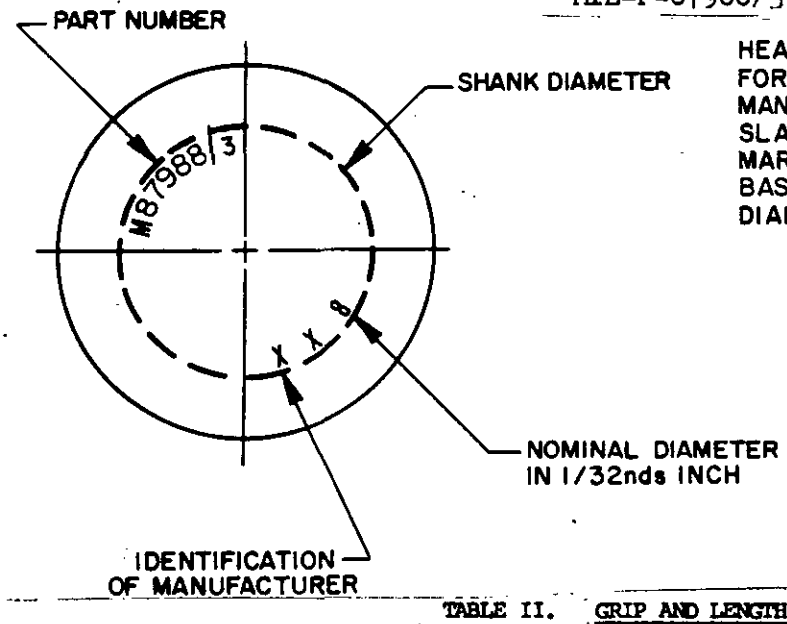
5. Part number. The part number consists of the letter "M", the basic number of this specification sheet, followed by a lubricant code letter, the first number indicates nominal diameters in thirty-seconds of an inch taken from TABLE I, the grip dash number indicates maximum grip length in sixteenths of an inch taken from TABLE II.

   Example: M87988/3B8-8



M87988/3B8-8 indicates: Pin-Rivet, Threaded, Type V (100° Flush MS24694 Tension Head), Class 1 (7075 Aluminum, 45 KSI Fsu and 75 KSI Ftu); anodized; solid film lubricant; 8/32 nominal diameter; 8/16 maximum grip length.

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HEAD MARKING SHALL BE .010 MAXIMUM DEEP. FOR DASH 5,6 AND 8 SIZES ONLY, MARK MANUFACTURER IDENTIFICATION AND M PLUS SLASH SHEET (M 3). FOR DASH 10 AND ABOVE, MARK MANUFACTURER IDENTIFICATION PLUS BASIC PART NUMBER (M87988/3-) AND DIAMETER.

TABLE II. GRIP AND LENGTH DIMENSIONS

GRIP DASH NO.	MAX GRIP ±.005	LENGTH ±.010[a/]						
		DIAMETER DASH NUMBER AND THREAD SIZE						
		5 .1640-32	6 .1900-32	8 .2500-28	10 .3125-24	12 .3750-24	14 .4375-20	16 .5000-20
1	.062	.375						
2	.125	.437	.450	.520	.625	.670	.760	.810
3	.188	.500	.513	.583	.688	.733	.823	.873
4	.250	.562	.575	.645	.750	.795	.885	.935
5	.312	.625	.638	.708	.813	.858	.948	.998
6	.375	.687	.700	.770	.875	.920	1.010	1.060
7	.438	.750	.763	.883	.938	.984	1.073	1.123
8	.500	.812	.825	.895	1.000	1.045	1.135	1.185
9	.562	.875	.888	.958	1.063	1.108	1.198	1.248
10	.625	.937	.950	1.020	1.125	1.170	1.260	1.310
11	.688	1.000	1.013	1.083	1.188	1.233	1.323	1.373
12	.750	1.062	1.075	1.145	1.250	1.295	1.385	1.435
13	.812	1.125	1.138	1.208	1.313	1.358	1.448	1.498
14	.875	1.187	1.200	1.270	1.375	1.420	1.510	1.560
15	.938	1.250	1.263	1.333	1.438	1.483	1.573	1.623
16	1.000	1.312	1.325	1.395	1.500	1.545	1.635	1.685
17	1.062	1.375	1.388	1.458	1.563	1.608	1.698	1.748
18	1.125	1.437	1.450	1.520	1.625	1.670	1.760	1.810
19	1.188	1.500	1.513	1.583	1.688	1.733	1.823	1.873
20	1.250	1.562	1.575	1.645	1.750	1.795	1.885	1.935
21	1.312	1.625	1.638	1.708	1.813	1.858	1.948	1.998
22	1.375	1.687	1.700	1.770	1.875	1.920	2.010	2.060
23	1.438	1.750	1.763	1.833	1.938	1.983	2.073	2.123
24	1.500	1.812	1.825	1.895	2.000	2.045	2.135	2.185
25	1.562	1.875	1.888	1.958	2.063	2.108	2.198	2.248
26	1.625	1.937	1.950	2.020	2.125	2.170	2.260	2.310
27	1.688	2.000	2.013	2.083	2.188	2.233	2.323	2.373
28	1.750	2.062	2.075	2.145	2.250	2.295	2.385	2.435
29	1.812	2.125	2.138	2.208	2.313	2.358	2.448	2.498
30	1.875	2.187	2.200	2.270	2.375	2.420	2.510	2.560
31	1.938	2.250	2.263	2.333	2.438	2.483	2.573	2.623
32	2.000	2.312	2.325	2.395	2.500	2.545	2.635	2.685

a/ Nominal length equals maximum grip plus "B". See Figure 1.

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TABLE III. INTERCHANGEABILITY

MILITARY PART NUMBER	HI-SHEAR EQUIVALENT PART NUMBER
M87988/3C5-( )	HL29-5-( )
M87988/3C6-( )	HL29-6-( )
M87988/3C8-( )	HL29-8-( )
M87988/3C10-( )	HL29-10-( )
M87988/3C12-( )	HL29-12-( )
M87988/3C14-( )	HL29-14-( )
M87988/3C16-( )	HL29-16-( )
M87988/3C indicates anodized and cetyl alcohol lubricant.	HL29 indicates anodized and cetyl alcohol lubricant.

Custodians:
Army - AV
Navy - AS
Air Force - 99

Review activities:
Army - GL
DLA - IS
NSA - NS

User activities:

Preparing activity:
Air Force - 82

Agent:
DLA - IS

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